Supporting Information for "A causal examination of the solar influence on Holocene climate"

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Introduction

This supplement provides material that supports the analyses in the main text. Section S1 contains additional information on embedding vectors, including an example illustrating how CCM skill varies with embedding parameters, and details about the CCM pipeline. Section S2 presents additional tests using alternative Greenland temperature estimates and δ^{18}O records, and GCM output from a transient Holocene simulation. This section begins by introducing supplemental data sets, then describes detrending considerations and finally, offers corresponding CCM results. Section S3 provides comparison of CCM results to those obtained using standard correlation. All figures and tables referenced in the main text appear here with their captions.

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1. SI: Text

Text S1: Method details

S1.1: Choice of embedding configuration

One of the cornerstones of CCM (and any state-space algorithm) is representing the state of the system at a given time point by using a snippet of recent history. Fig. S1 illustrates how the state associated with one time point can be described in various ways depending on choice of E (embedding dimension, or number of points in the snippet) and τ (time steps between those points). The combinations of (E, τ) yield determine the length of time and resolution of the snippet (embedding vector). The art of choosing E and τ is in capturing the most informative window at a level of detail that neither over values noise, nor loses signal to smoothing. To determine an appropriate choice of E and τ, it is common to evaluate self-prediction using the simplex method (Sugihara et al., 1997) and choose a common configuration that yields high skill. In this case, no such optimal pair emerged. Fig. S2 shows simplex self-prediction results that motivate our parameter-sweep approach.

S1.2: CCM steps

We use CCM to analyze the relationship between temperature and TSI across four dyads, representing all combinations of two temperature and two TSI reconstructions. As necessary, time series were averaged to a common decadal time axis. Broadly, the steps of CCM are as follows for testing the relationship, "TSI influences temperature":

Step 1: Construct time delay embeddings for each variable (e.g. TSI and temperature) using specified values of E (embedding dimension) and τ (time delay).

Step 2: Randomly sample m GMST embedding vectors to build a library of example states,

Step 3: Identify the k nearest neighbors from the library to each index j in GMST, excluding vectors within a radius of $(E-1)\times\tau$ to avoid dependent samples.

Step 4: Estimate the value of TSI at $t=j+1$ by averaging the values in TSI corresponding to $t=i+1$ for indexes corresponding to nearest neighbors.

Step 5: Calculate CCM skill (ρ) as the correlation between observed and predicted values of TSI.

Step 6: Repeat n times across each of a range of library sizes, m, to generate a representation of the relationship between CCM skill and library size.

Calculations were run based on the pyEDM software (Park, 2024).

In this study, we used 20, rather than the typical 5 nearest neighbors. This is a conservative choice to reduce the probability of spurious matches. Additionally, vectors within a radius of $(E-1)*\tau$ were excluded from the pool of nearest neighbor candidates to avoid dependent samples.

For each E-τ configuration, we evaluated CCM skill across a range of relative shifts between the two series. In CCM, the strongest reconstruction typically occurs when the predictor is shifted forward to align the forcing with its delayed expression in the outcome. Accordingly, we scanned lags for the "TSI influences temperature" direction and report final results for the positive shift (TSI precedes temperature) that yields the highest convergent predictive skill for each configuration. Given inherent age uncertainty in the data, we additionally allowed lags whose identified peak spanned zero. This approach is somewhat simplistic, as it assumes that the process of imprinting influence on archive is constant, but is more logical than assuming that the imprint will be immediate.

Figure S3 shows library-size-ρ trajectories for a subset of time series alignments (ℓ) for the ($E = 4$, $\tau = 1$) configuration of the ALLEY00-VIEIRA11 dyad. Figure S4 shows how these results are summarized in the ($E = 4$, $\tau = 1$) subplot, the basis for arriving at an optimal ℓ, and how different E-τ pairs result in different optimized (E, τ, ℓ) combinations. These figures illustrate the calculations behind the summary statistics in the main-text results (Figure 3) and S6 (among others) and do not introduce results beyond those discussed in the manuscript.

S1.3: Surrogate testing

To evaluate the significance of causal relationships (for a given E-τ-ℓ configuration), we compare CCM output generated evaluating the relationship between real variables to that generated when one variable is replaced by a set of surrogates. Here we used phase-randomized surrogates per Prichard and Theiler (1994), generated using Pyleoclim (Khider et al., 2022). Phase-randomized surrogates maintain the original autocorrelation structure (hence: power spectrum) of a series while randomizing its phase, thereby breaking any temporal connection to the other relationship variable, but preserving the possibility of observing similarities by coincidence. A relationship is considered significant if the mean value of the final 30 library sizes of the real relationship is greater than 95% of comparable values calculated from each type of surrogate relationship (200 each using either TSI or temperature surrogates). Cases failing this significance criterion are highlighted with semi-circles sized according to the degree of failure, showing where surrogate tests undermine the original relationship (e.g., Figure 3 (main text) and S6).

Text S2: Supplemental CCM calculations

S2.1: Data sets

Ice Core Temperature Reconstructions: The δ^{18}O-derived temperature estimate from GISP2, publicized by Alley (2000), is widely used in Holocene climate studies because of its length, continuity, and annual chronology based on replicated layer counting. Subsequent work by the ice core community has yielded alternative age models (e.g., GICC05 (Svensson et al., 2008) and GICC21 (Sinnl et al., 2022) chronologies) and additional temperature reconstructions, including firn-diffusion inversions. To evaluate whether our CCM results depend on the choice of Greenland record, we expanded the analysis to include reconstructions that differ in chronology, reconstruction method, and site.

Döring and Leuenberger (2022) (Fig. S5d) infer surface air temperature at GISP2 by inverting a firn-diffusion model to fit δ^{15}N observations. The inversion adjusts annual temperature and accumulation so the forward model matches the measured δ^{15}N as closely as possible. We use their reconstruction based on the Goujon, Barnola, and Ritz (2003) formulation, identified as the most robust. All estimates follow the GICC05 (Svensson et al., 2008) timescale.

Martin et al. (2024) (Fig. S5e, f (GISP2, NGRIP)) estimate temperature for GISP2 and NGRIP using a coupled firn-densification and heat-diffusion model that evolves the firn column forward through time. Temperature and accumulation are constrained by model physics rather than tuned to fit δ^{15}N. Their reconstructions follow a bipolar chronology that synchronizes WD2014 (Antarctica, (Buizert et al., 2015)) and GICC05 (Greenland, Svensson et al. (2008)) using methane and volcanic tie points.

Seierstad et al. (2014) (Fig. S5g, h (GISP2, NGRIP)) present δ^{18}O records for GISP2 and NGRIP, adding targeted measurements and using volcanic tie points and other stratigraphic markers to place both cores on the GICC05 timescale (Svensson et al., 2008).

GCM output

Tian, Jiang, Zhang, and Su (2022) (Fig. S5a) published results from a transient Holocene simulation run with the "ALL" CESM configuration. This configuration, forced with constant solar luminosity but variable insolation, CO_2 and freshwater forcing, provides a counterfactual for the spurious detection of TSI influence.

S2.2: Procedural considerations

Because CCM relies on pattern matching, underlying linear trends or long period variations can confound the algorithm. This raises a challenge, however, because in nonlinear systems, it is uncommon to encounter a truly secular and known underlying condition. The task of removing such confounding macrostructure is fraught because–especially in our case, which seeks to characterize a subtle response to a weak forcing–even small changes in the time series may make a pair of time series snippets appear artificially analogous or artificially distinct, and subsequently distort the assessment of influence. Several of the additional ice core records and the GCM output exhibit low-frequency drift so we tested two treatments: linear detrending and applying a 1kyr high-pass filter. Linear detrending is among the most minimal interventions for non-stationarity. It has little effect on records such as Seierstad et al. (2014) or Alley (2000) GISP2, which were already close to stationary, but removing trend from the Döring and Leuenberger (2022) GISP2 reconstruction and Seierstad et al. (2014) NGRIP δ^{18}O record revealed previously muted influence. The 1kyr high-pass filter was motivated by concern that long-period variation in insolation might inflate the predictability of temperature timeseries independently of TSI, and the expectation that the climate response would be generally sub-millennial. Because longer-period solar variability (e.g., the Hallstatt cycle at $\sim$2,400 years) may contribute genuine signal, we interpret the filtered results as a conservative lower bound on the evidence for influence.

Fig. S5 presents the raw, linearly-detrended, and 1kyr high-pass filtered versions of each record (original and supplemental), and associated PSD plots.

S2.3: Results of supplemental analyses

Fig. S6 extends the main text Fig 3 to include results for dyads relative to WU18 for each of the additional records described in S2.1, using their untreated, linearly detrended and high-pass filtered forms. Because of the potential for mathematical artifacts discussed in section S2.2, we interpret the extended results at a relatively high level. TSI had a broad, discernible influence on untreated dyads involving ERB22, and the Seierstad et al. (2014)

and Alley (2000) GISP2 records. Upon linearly detrending, some previously significant configurations became insignificant, while influence is newly detected among Döring and Leuenberger (2022) (GISP2) and Seierstad et al. (2014) (NGRIP) configurations. In many cases, high-pass filtering reduced the number of significant configurations. The notable exception was the Martin et al. (2024) (NGRIP) dyad, which only yielded configurations with significant influence after high-pass filtering. Collectively, this suggests that the main-text finding that hinged on ALLEY2000–that TSI exerted a discernible influence on Greenland climate as recorded at these sites over the preindustrial Holocene–is not sensitive to site, age model, or observation type. The pattern of null results among the Martin et al. (2024) reconstructions may reflect their stronger physical constraints, which smooth or reorganize the fine-scale structure that CCM relies on, rather than the absence of a solar imprint.

Almost all configurations of the Tian et al. (2022) dyad–regardless of treatment– returned insignificant results and low $\Delta\rho$ values, providing a negative test of whether CCM is liable to return false evidence of influence.

Text S3: Comparison to a linear model

The 11-year Schwabe cycle provides an empirical demonstration that TSI and temperature are directly related on short timescales (Amdur et al., 2021). It is therefore worth asking whether the longer-timescale relationship we detect with CCM could be recovered by a simpler linear approach. To that end, we replaced CCM with standard correlation and followed our pipeline as closely as possible; for a given dyad, we calculated correlation between the two time series at the resolution determined by τ over the same range of lags, and tested against phase-randomized surrogates in the same style. For each treatment of each dyad we report the correlation at the optimized lag, alongside the fraction of surrogates that outperformed the real relationship, for each value of τ.

Broadly, dyads were positive correlation at their optimal lag, but that correlation was insignificant in most cases when tested against surrogates (Fig. S8). Those configurations that were significant were sparsely distributed such that no dyad showed consistent significance regardless of treatment. In most cases, correlation decreased after high-pass filtering, while the fraction of surrogates that outperformed the real timeseries increased.

The color in Fig. S10 is calculated as the difference between CCM skill at each $E - \tau$ configuration (seen in Fig. S9) and the simple correlation for the corresponding τ value (per Fig. S8). The difference between CCM skill and simple correlation indicates that in the majority of cases the relationship is better described in state space rather than in time.

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2. SI: Figures

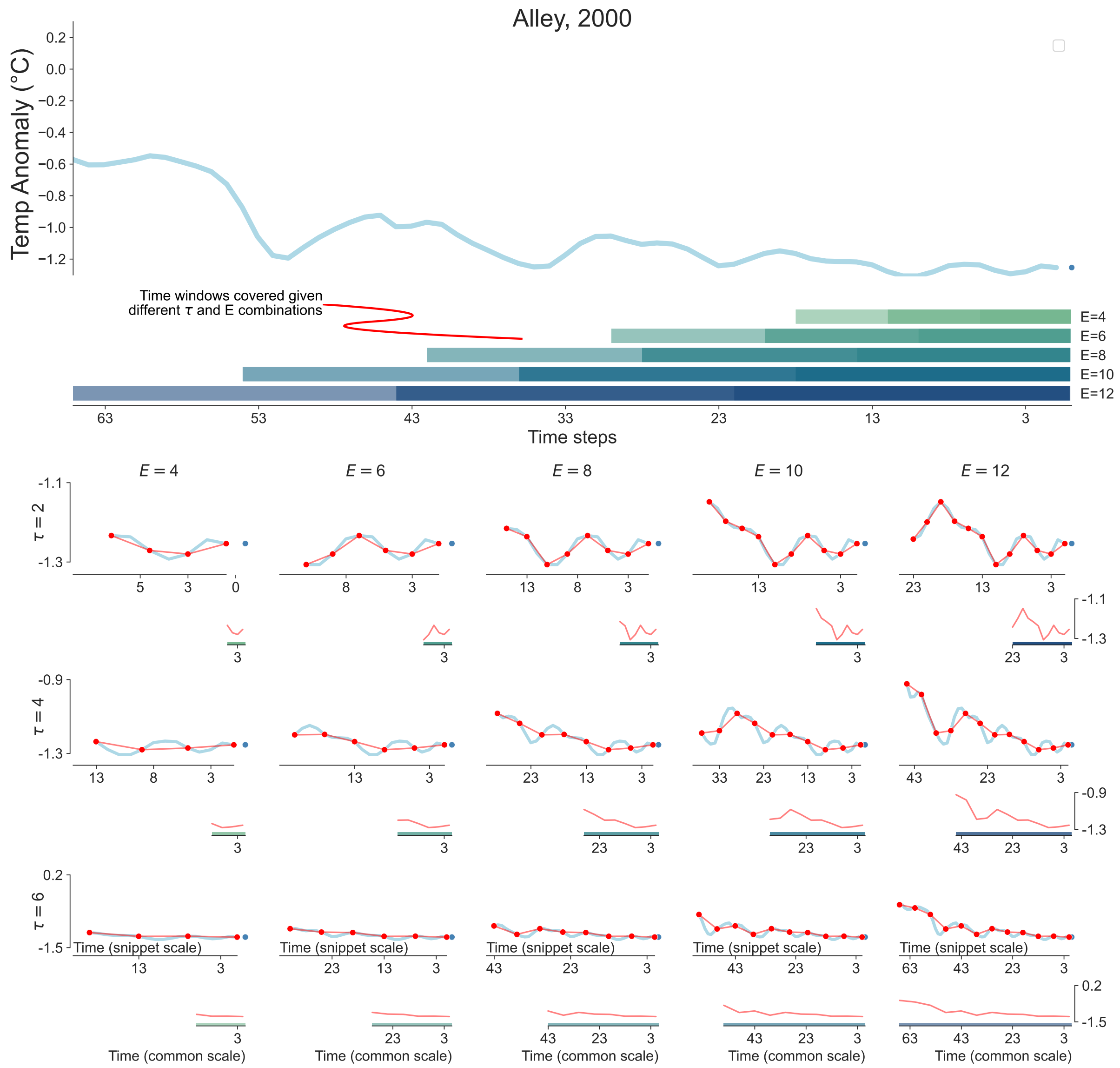

Figure S1. Choosing E and τ. Illustration of how embedding parameters determine the snippet used for reconstruction. The top panel [line plot] shows a segment of the GISP2 record leading up to a target point. The middle panel [bar plot] shows how different choices of E and τ define snippet length: for each E (distinguished by color), longer bars correspond to larger τ. The lower panel [multiplot] shows examples of these snippets, plotted both in their native time span and on a common scale. Collectively, these plots highlight how different choices of E and τ change snippet appearance.

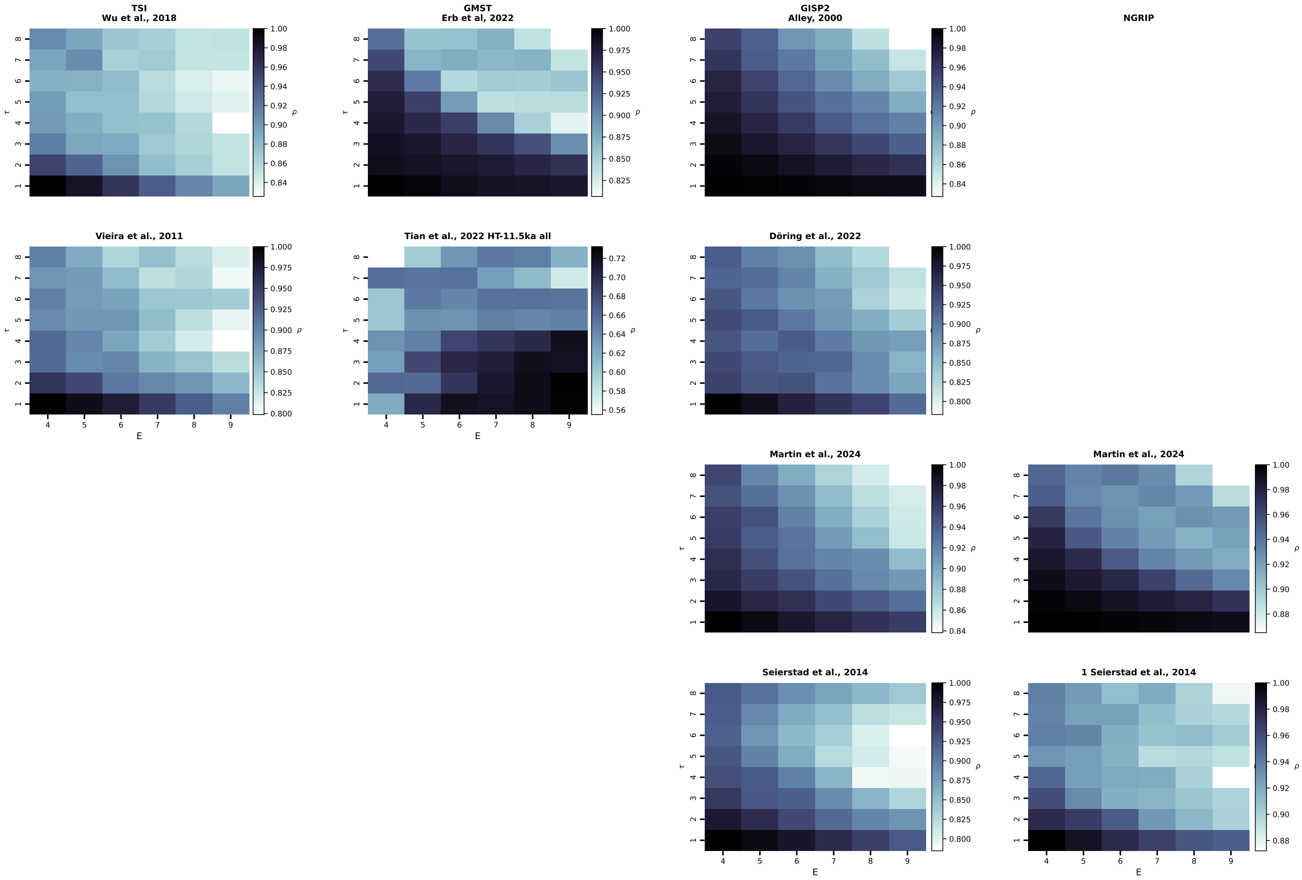

Figure S2. Determining the embedding configuration. Each grid cell shows the simplex (Sugihara & May, 1990) self-prediction skill for a given time series across combinations of embedding dimension, E (column) and τ (row). Color bars represent the prediction accuracy on a scale of [0-1]. Note that the minimum and maximum values for each color bar are variable specific. Differences in the patterns of self-prediction between TSI time series and climate time series motivated the decision to pursue a range of E-τ configurations with CCM.

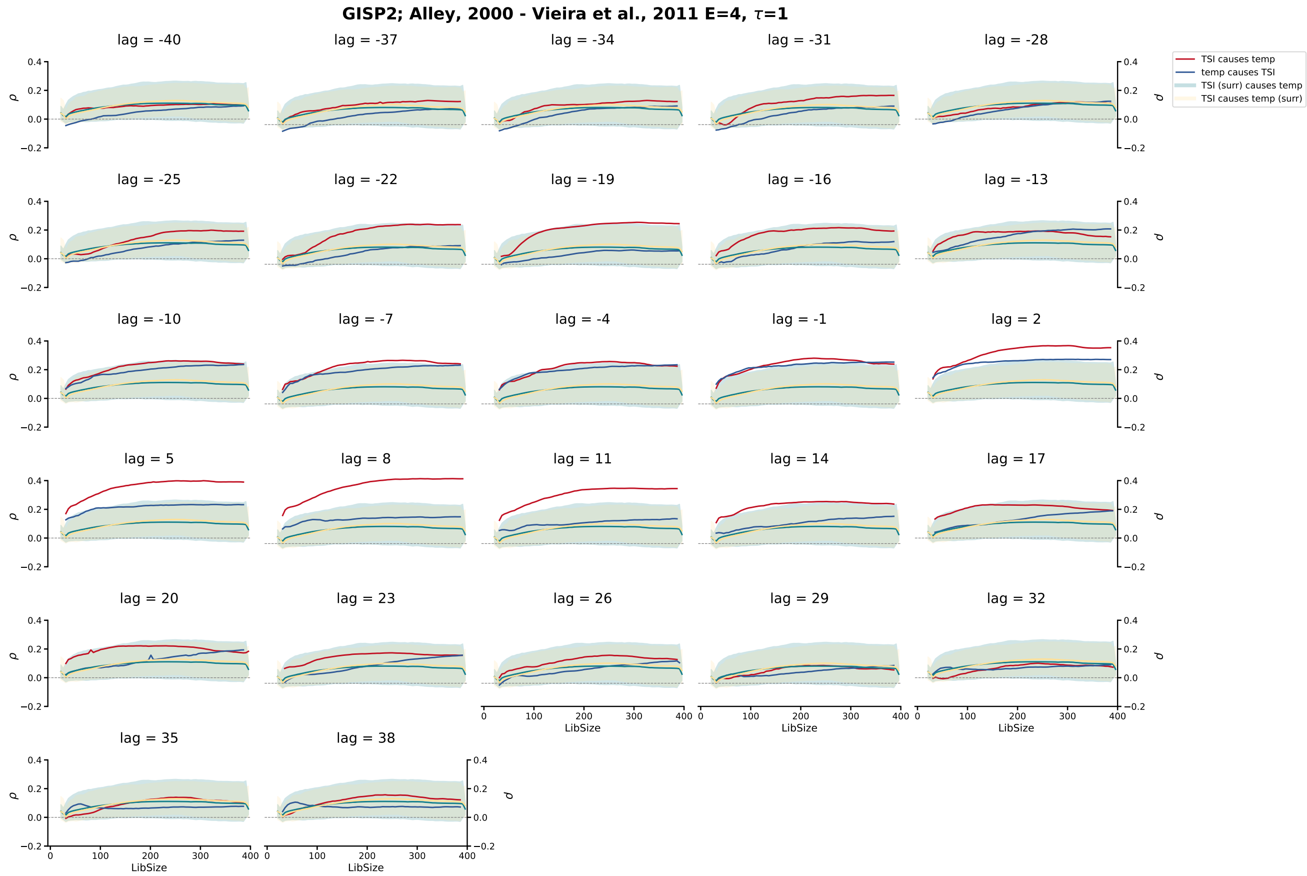

Figure S3. Library size–ρ trajectories for the ALLEY00-VIEIRA11 dyad at ($E = 4$, $\tau = 1$). Each subplot corresponds to a different alignment of the two time series. Lines show "TSI influences temperature" [red] and "temperature influences TSI" [blue], with shaded envelopes indicating the 5th–95th percentile ranges of the corresponding surrogate null distributions ("TSI (surr) influences temperature" [green] and "TSI influences temperature (surr)" [yellow]). Superior performance by the "TSI influences temperature" relationship at positive lags is consistent with a realistic physical link, since shifting TSI forward for optimal performance aligns with the expectation that TSI leads temperature.

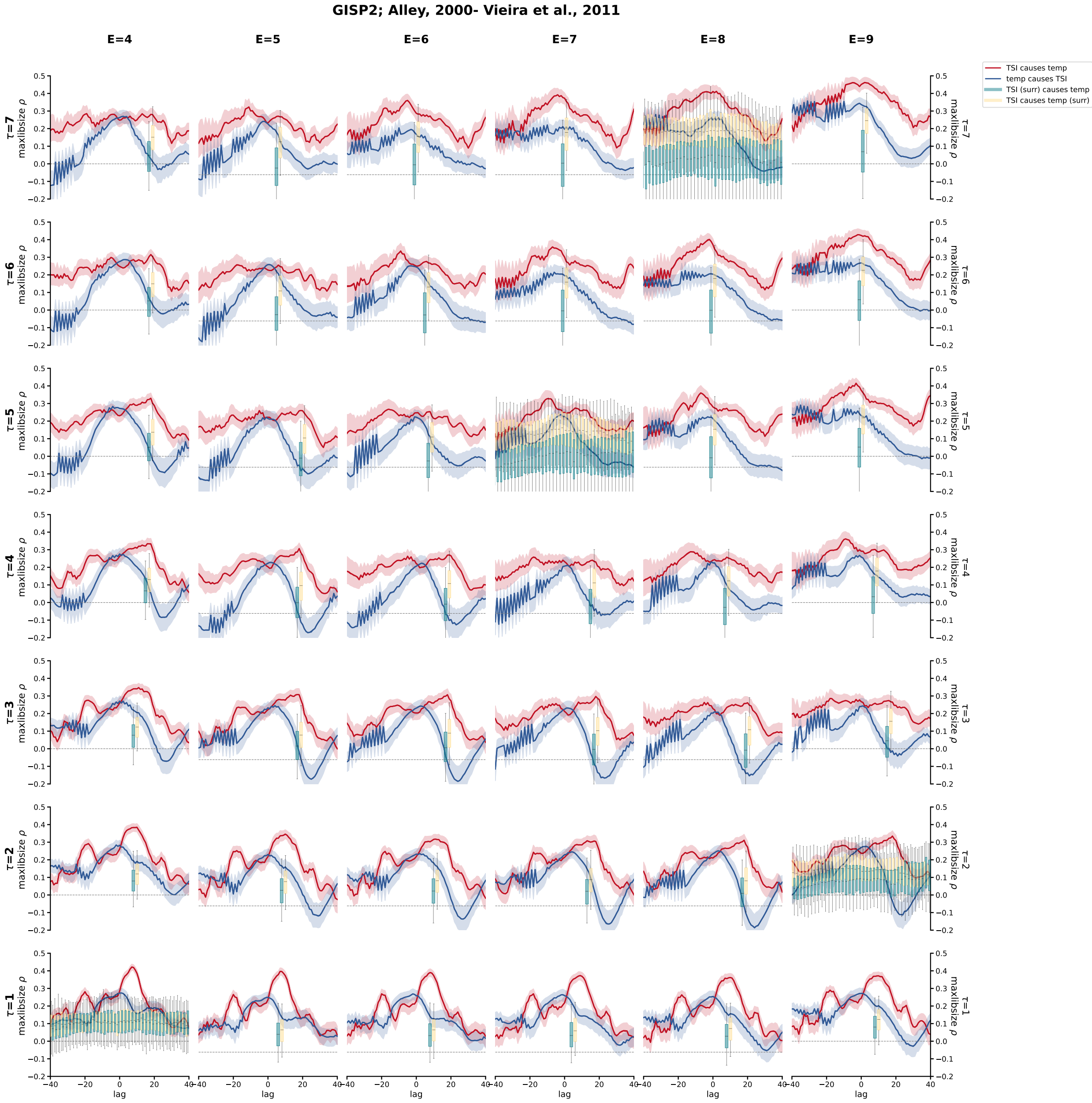

Figure S4. Final CCM skill ($\rho_{\mathbf{final}}$) per lags for (E, τ) configurations; ALLEY00-VIEIRA11 dyad Each subplot corresponds to one (E, τ) configuration and shows the ρ_{final} values across lags for the real relationships ("TSI influences temperature" [red], "temperature influences TSI" [blue]). Surrogate null distributions ("TSI (surr) influences temperature" [green] and "TSI influences temperature (surr)" [yellow]) are summarized by boxplots (boxes: inter-quartile range, whiskers: 5th–95th percentiles). The optimal lag is the shift with the highest convergent ρ_{final}, and is significant if this value exceeds the 95th percentile of the better-performing surrogate class. Because phase-randomized surrogates preserve spectral content but not alignment, their distributions are invariant to lag; after confirming this in a subset of cases, we restricted surrogate testing to a single lag.

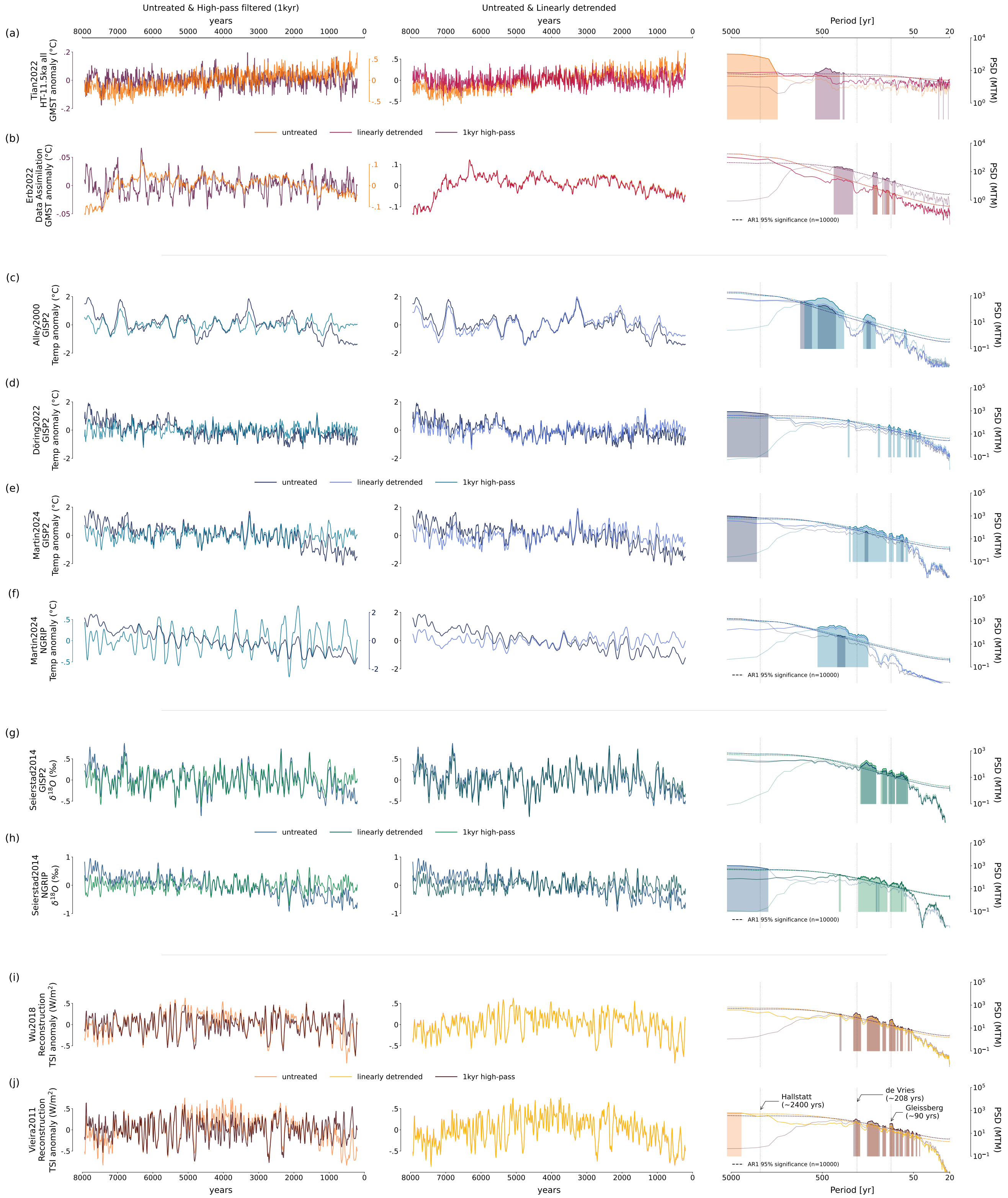

Figure S5. Raw and detrended timeseries Each row corresponds to a source timeseries. The left column plots compare the untreated and 1kyr high-pass filtered versions, the middle column compares the untreated and linearly detrended versions, and the right column shows power spectral density of all three time series versions with AR1 significance testing.

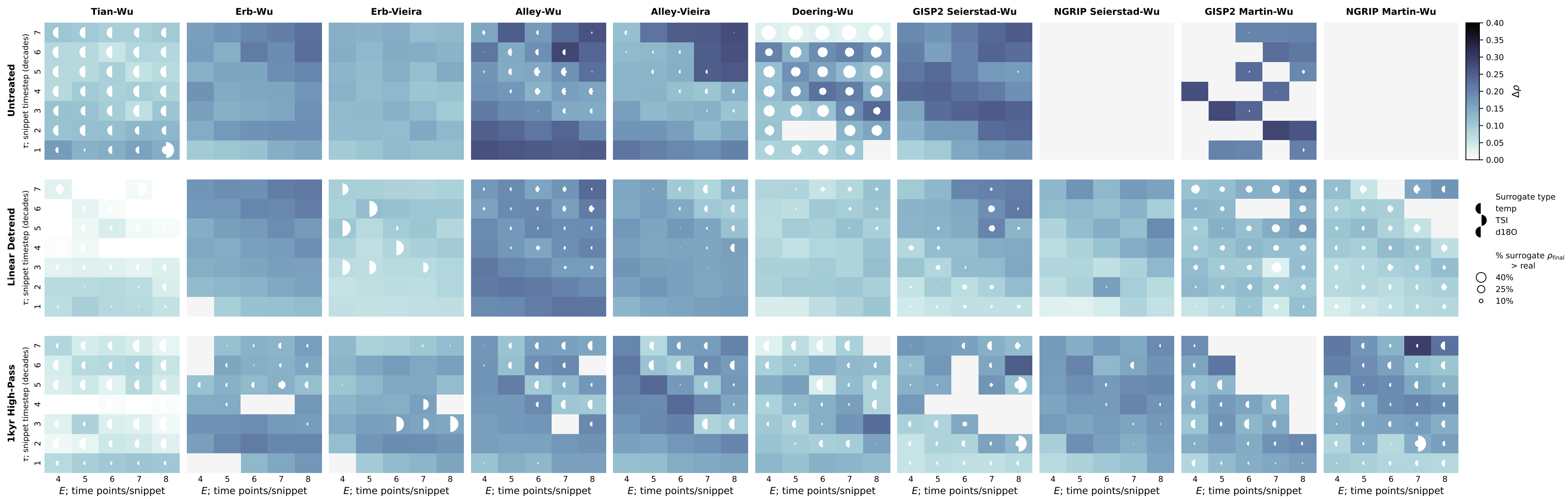

Figure S6. **Full result overview.** Each grid summarizes CCM skill ($\Delta\rho$) and significance for the relationship "TSI influences temperature" of a particular dyad across various embedding combinations, E as columns and τ as rows. Half-moons annotations indicate surrogate failure rates (percent of surrogate relationships with final $\rho \geq$ real final ρ); left half-moons for "TSI influences temperature (surrogate)" and the right for "TSI (surrogate) influences temperature". Within each subpanels, columns correspond to climate records, with analysis of the untreated timeseries in the top grid (original) and analysis of the linear detrended version in the bottom grid. Cells without a viable lag candidate are masked with light gray.

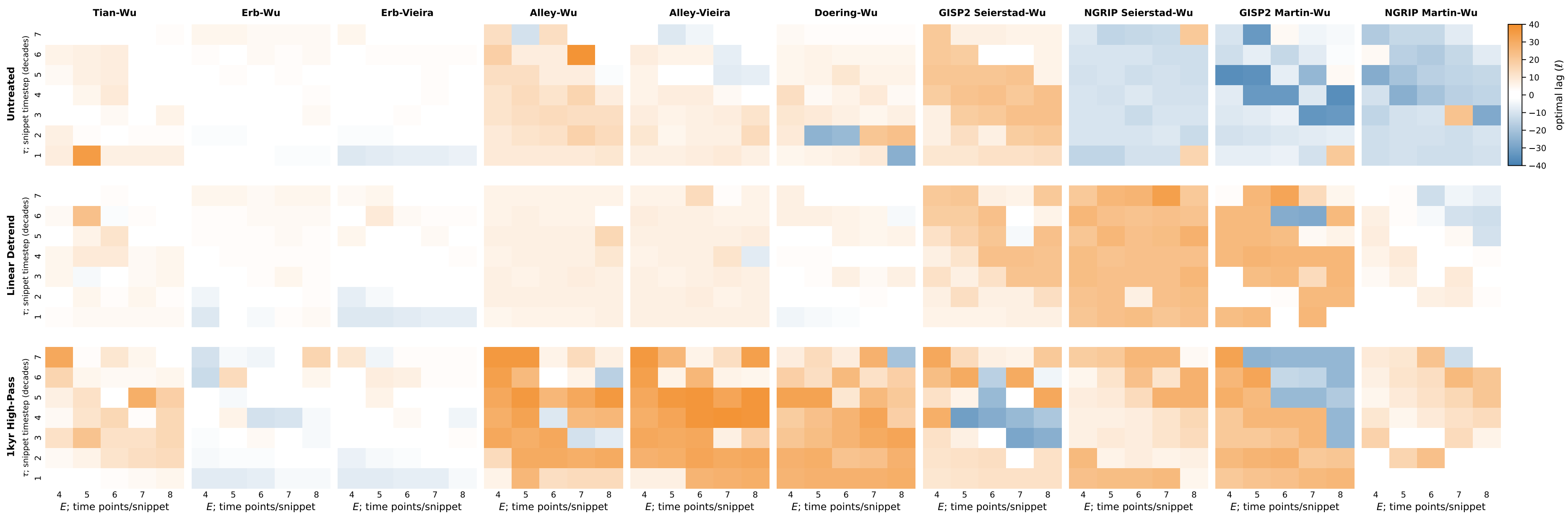

Figure S7. **Lag (ℓ) choice** Choice of lag for reported results for each dyad-treatment-$E-\tau$ configuration.

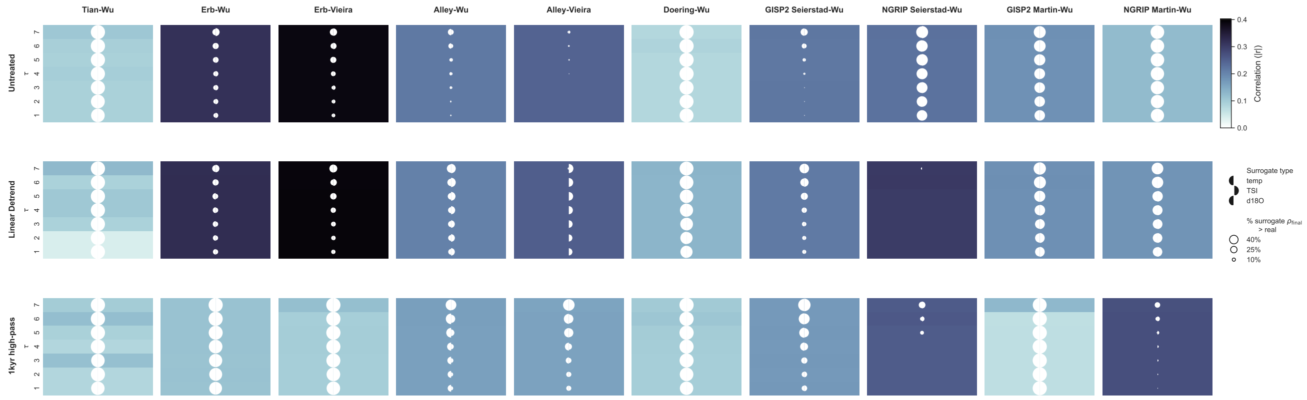

Figure S8. Simple correlation of all dyads Each subplot reflects a dyad-treatment combination (e.g. Erb-Wu, untreated). The columns of the subplot grid correspond to dyads, and rows to timeseries treatment. Within each subplot, blocks are organized vertically per τ and their color corresponds to the optimal correlation obtained in a lag scan (per the CCM pipeline) and half moons indicating fraction of surrogates outperforming real result (left half moon: temp and $\delta^{18}O$, right half moon: TSI).

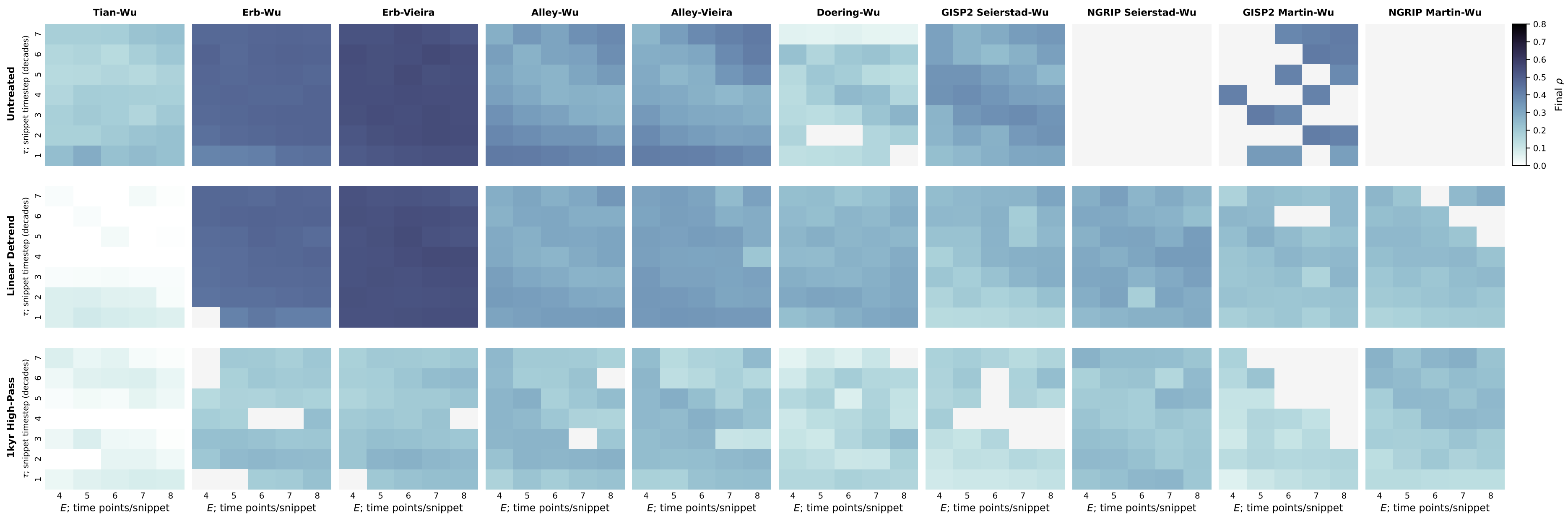

Figure S9. **CCM skill for optimal lag** This plot is organized in the same way as Fig. S6. Here, color corresponds to CCM skill after convergence (ρ at largest library size).

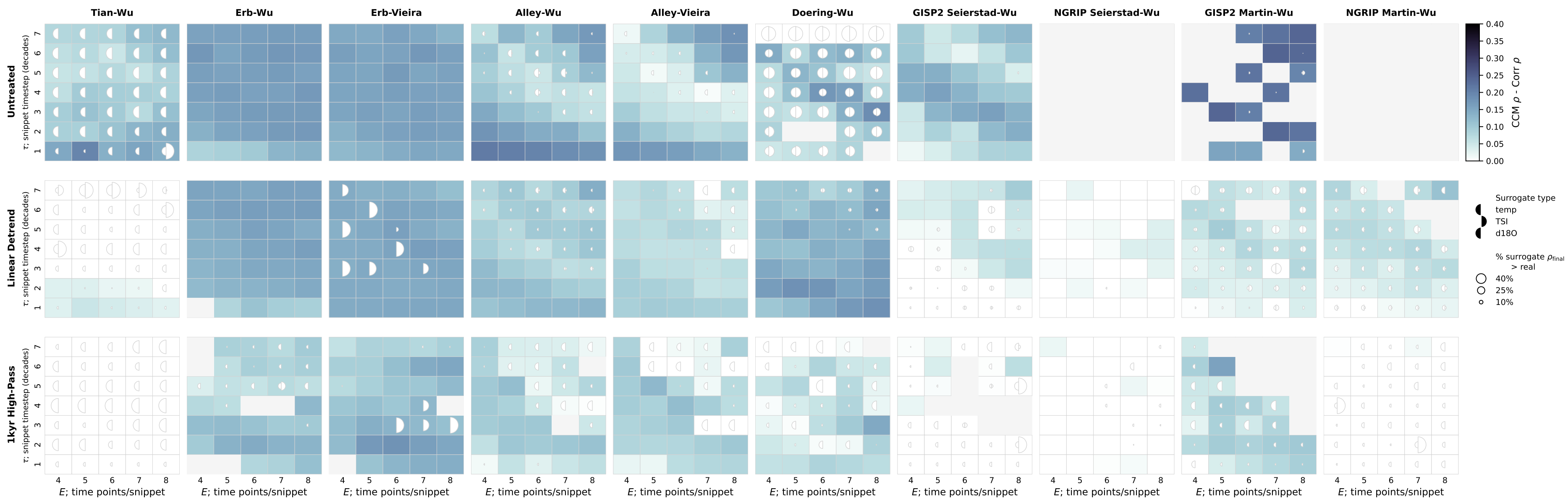

Figure S10. Difference between CCM skill and simple correlation This plot is organized in the same way as Fig. S6. Here, color corresponds to the difference between CCM skill (ρ at largest library size) and the simple correlation calculated for the dyad/treatment combination for the shared value of τ.